

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

AI in Cyber Security [IT_4004]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Using a suitable example, explain how **supervised and unsupervised learning algorithms** can be applied in cybersecurity for **anomaly detection** and **threat classification**. Illustrate the process with a labeled diagram or workflow. (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) Design a **use case scenario** where **machine learning techniques** enhance cybersecurity-such as spam filtering, intrusion detection, or ransomware detection. Outline the dataset features, model type, and expected outcomes. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) **Problem – Email Spam Detection** (5)

A)

Email	Contains Link	Contains Offer	Spam
1	Yes	Yes	Yes
2	Yes	No	Yes
3	No	Yes	No
4	No	No	No
5	Yes	Yes	Yes
6	No	No	No

I. Calculate the Entropy of the dataset (target: Spam).

Use the formula:

$$H(S) = -p_{Yes} \log_2 p_{Yes} - p_{No} \log_2 p_{No}$$

II. Compute the Information Gain for:

1. Contains Link
2. Contains Offer

$$IG(S, A) = H(S) - \sum_{v \in A} \frac{|S_v|}{|S|} H(S_v)$$

III. Decide the Root Node for the Decision Tree.

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) Compare and contrast **signature-based** and **behavior-based malware detection** approaches. (5)
Analyze which method is more effective in detecting **zero-day malware** and justify your reasoning.

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Analyze the impact of **feature selection** and **dimensionality reduction** (e.g., using PCA) on the accuracy and interpretability of ML models used in cyber threat detection. Provide a suitable justification with examples. (5)

A)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) Discuss how deep neural networks (DNNs) improve accuracy and adaptability in intrusion detection systems (IDS). Illustrate with an example from recent cyberattack scenarios. (5)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) Examine a case study on detecting DDoS or brute-force attacks using LSTM or CNN models. Discuss model design, dataset used, and results obtained. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Compare the performance and characteristics of traditional machine learning (e.g., SVM, KNN) and deep learning models (e.g., CNN, LSTM) for intrusion detection. Analyze which approach is more effective for large-scale, high-dimensional network datasets. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5) Discuss how **GAN-based adversarial attacks** exploit the interaction between the **generator** and **discriminator** networks. Analyze why GAN-based attacks are considered a major threat to **image classification and deepfake generation**. (5)

A)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) Explain how an adversary can manipulate training data to perform a poisoning attack. Differentiate between dataset poisoning, algorithm poisoning, and model poisoning with suitable cybersecurity examples. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	15
3	3	CO 3	15
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
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1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	20(40%)
4	Analysis	6	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)